The TRON Church

Central AV Team Training Session

Sound

Name:

THE TRON CHURCH 25

Why?

"…through love serve one another"

Galatians 5:13

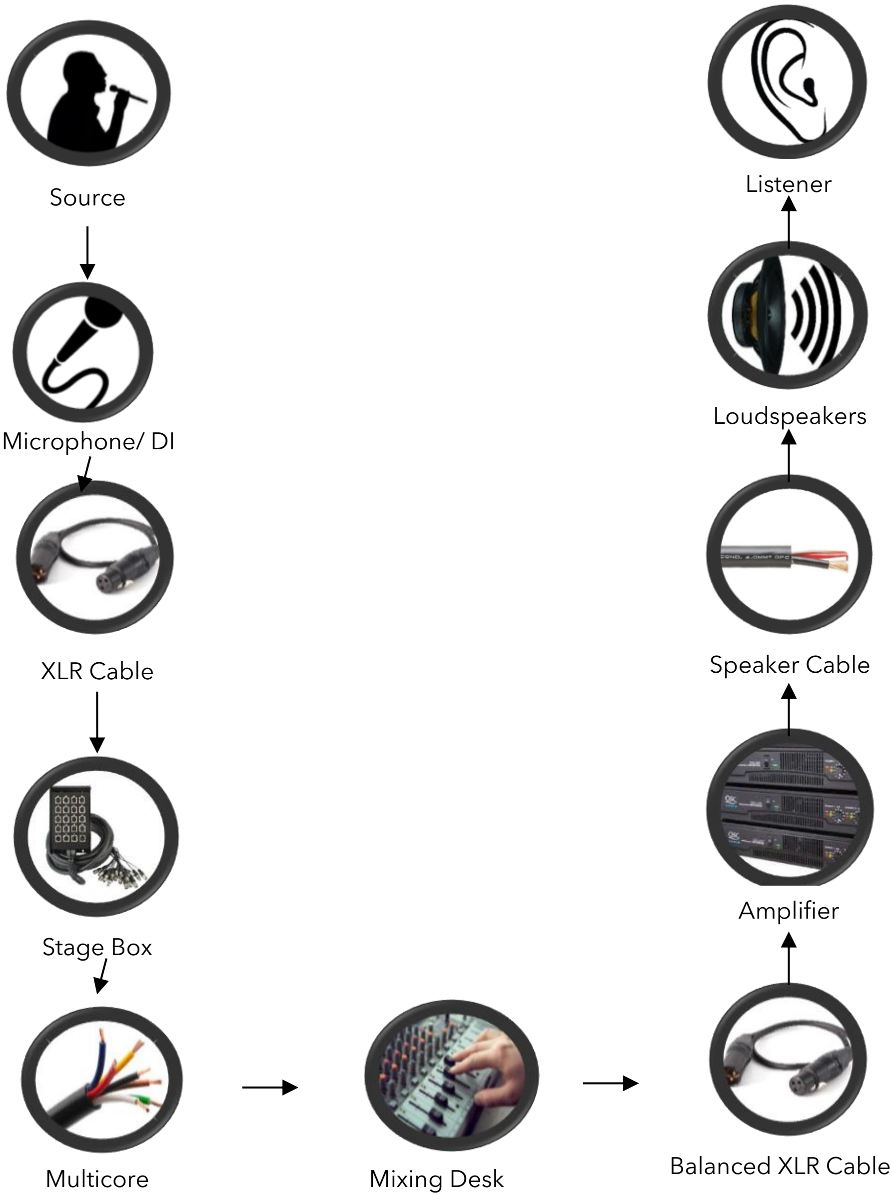
To facilitate the preaching of God’s word and praising of His name through excellence in AV so that people will hear, participate in and be RENEWED in our 3 locations and online.

Notes:

What?: Loud and Clear

How?: Sound to Source- A Basic Overview

This diagram shows the outline of the journey the audio signal takes from source to listener.



	**SOURCE** The source is whatever or whoever is producing the sound you wish to capture. For our purposes this normally someone speaking, singing, or a musical instrument.
	**MICROPHONE/ DI BOX** Microphones are used to capture the sound from the source and turn the sound into an electrical signal ready to be sent to the mixing desk. The placement of the mic is important and will vary depending on the source you wish to capture. More on this later… DI boxes or Direct Injection boxes convert the high impedance and unbalanced signals from electronic instruments such as guitars or keyboards into a low impedance, balanced signal which is suitable for the mixing desk.
	**XLR CABLE** An XLR cable carries the signal from the microphone or DI box to the stage box (or in some cases straight to the mixing desk if the desk is near to the band). The XLR cable is ‘balanced’ which means it can carry very low level signals (such as those from a microphone) without picking up any unwanted noise.

 	**STAGE BOX AND MULTICORE** The stage box is like a hub for all the XLR cables from microphones and DI boxes to plug into on the 'stage' (band) end. The multicore then carries all of these signals in one thick cable to the mixing desk. This is a neater and more efficient way of connecting multiple XLR sources to the mixing desk.
	**MIXING DESK** The mixing desk is the means by which we get the final sound or 'mix' that we want to send out. It takes all of the signals from the different sources and allows us to 'mix' them together and adjust the volumes independently.
	**BALANCED CABLES** The 'mixed' signal then travels from the mixing desk to the amplifiers. In some cases the output will also be sent back down the multicore so that feedback monitors can be connected near the band
	**AMPLIFIERS** Amplifiers amplify (make louder!) the mixed signal from the mixing desk. Some systems will have 'active' speakers which house the amplifiers inside the speakers themselves

 	**SPEAKER CABLES AND SPEAKERS** The amplified signal is then carried to the speakers via speaker cables and the signal is reproduced by the speakers in the room

How?: Microphones

What is a microphone?

- A microphone is a transducer
    - A transducer is something that converts one from of energy into another. Microphones convert sound energy into electrical energy.

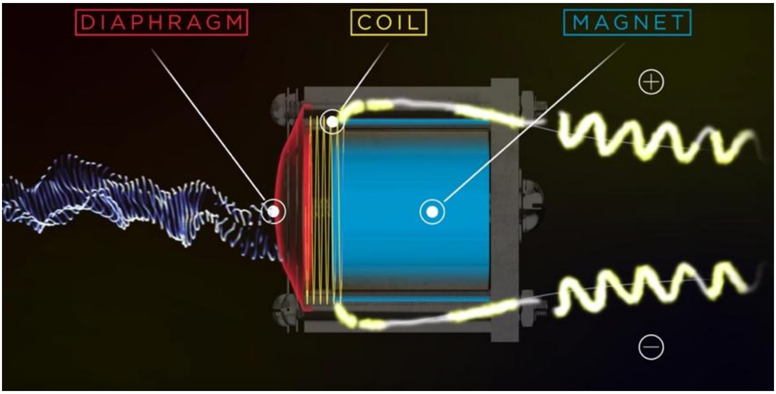
The main types of microphones

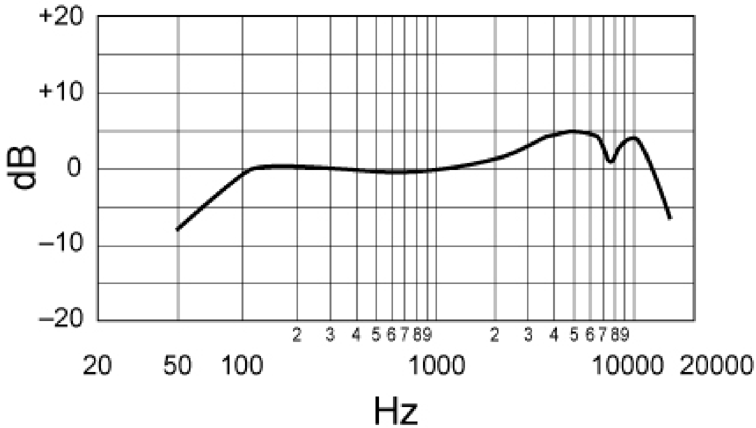
There are two main types of microphones which we are likely to use in a church setting:

Dynamic (also called 'moving coil')

- Dynamic microphones have an uneven frequency response as the diaphragm of the microphone is moved more or less depending on the frequency as shown below.
- This can be advantageous as they will naturally eliminate high frequencies and boost frequencies around the peak at 4-5kHz.

High frequency signals are unable to move the heavy coil of the mic so the response falls off. This makes them useful for singers as they will naturally boost the vocals of most singers and simultaneously cut higher frequencies which gives presence to the voice.


DIAPHRAGM
COIL
MAGNET


+20
+10
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dB
2 3 4 5 6 7 8 9
2 3 4 5 6 7 8 9
20
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Hz

SM58

SM57

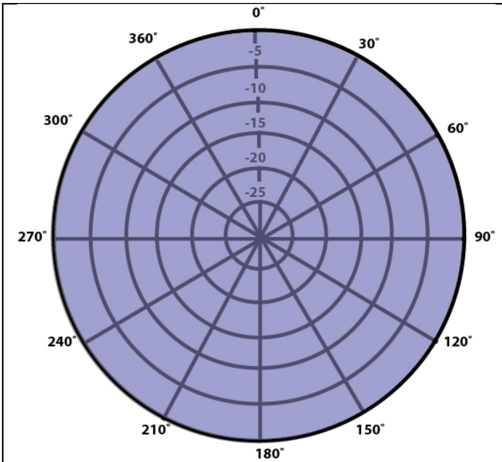
The SM58 and SM57 microphones pictured above are common examples of Dynamic microphones. SM58s are particularly useful for singing and SM57s are useful for wind instruments.

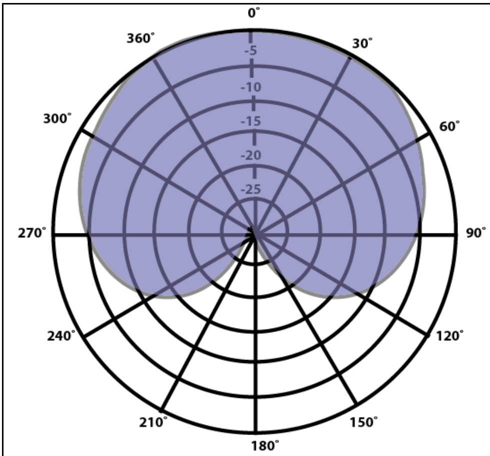
Condenser (also called 'capacitor)

The diaphragm of a condenser mic is not attached to a heavy coil as in a dynamic mic. This means that the frequency response is more flat and the microphone is more responsive to high frequencies. This means they give a more natural sound than a dynamic mic which boosts certain frequencies. These microphones are very popular for recording classical music and are particularly useful for picking up classical stringed instruments.

Understanding Microphone Polar Patterns

For our purposes, there are two types of polar patterns which we need to think about. They are shown in the table below. The 3D graphic to the right may help in understanding the diagrams in the table.

	**OMNIDIRECTIONAL** Omnidirectional mics will *theoretically* pick up sound from all directions. Generally, tie clip or headset microphones are omnidirectional. This pickup pattern is preferable as it means the microphone can be placed without having to worry about the speaker moving their head.

	**CARDIOID** Cardioid mics will pick up most sound from the front and slightly reduced output from the sides and *theoretically* zero from the rear. This makes them useful for instruments and singers as they can be directed at a single source without picking up other interfering sounds. **SM58s, SM57s and the pencil mics on the pervious page are all forms of cardioid mics.**

Microphone Placement:

The correct placement of the microphone depends on the nature of the source you wish to pick up as well as the surroundings and potential interfering sources nearby.

Ideally, when we have multiple musicians, **each microphone will only pick up sound from one source**. This allows us to isolate each sound and balance them the way we want to through the mixing desk.

If microphones are picking up sound from other sources then the sound mix may sound a bit 'muddy' and unclear.

When placing microphones, bear in mind the polar patterns on the previous page. If you want to reject an unwanted sound, point the microphone away from it and directly towards the intended source.

The distance and position of the mic in relation to the instrument varies from instrument to instrument. Often the musicians themselves will know the best place to put it so communicate with them. Also ensure when placing microphones that the mic and stand are not in the musician's way. Once you have positioned the microphone it is a good idea to check it's okay with them.

The screenshot above is an example of good mic placement. Matt's microphone is at the correct height and is close to his mouth. The mics for the violins and viola are pointed directly at the bridge of the instrument they want to pick up and Beccy's microphone is, as far as possible, pointed away from the piano to avoid unwanted interference.

Headset Mic Placement:

Getting the headset mics adjusted and fitted properly makes our job on the sound desk easier. The picture below is a good example of where the microphones should be placed in relation to the preacher's mouth.

Note that the microphone is not protruding beyond the mouth of the speaker and it is not sticking out far at the side. This will make it easier to get a good sound and will also make it more comfortable.

Getting a good fit is obviously down to the wearer more than us but it is helpful if we communicate with those speaking and advise them how it should be fitting, particularly if they are a guest speaker or if it's someone who is not used to using headset microphones.

The diagram below shows how to adjust the mics we use.

Priorities:

The single **most important mic** in any of our services is the preacher's mic. We want them to have full confidence that when they stand up, the mic will be working and sounding clear.

Therefore, **test this mic thoroughly(!).** Put the headset on the way the preacher will; Project your voice when testing; Don't try to shout over the musicians as they rehearse, wait for a time when you have some quiet to test thoroughly.

Mixing Desk: Volume

When setting volume, we have two main tools to use – Gain and Level (the slider).

Gain is adjusted using a dial normally near the top of the desk and level is adjusted using the sliders normally at the bottom.

Both affect the volume but, in simple terms, gain refers to the input of an audio signal where level refers to output.

If you are finding that you have the slider right up to the top but it's still not loud enough, chances are the gain is too low and needs to be increased.

Be careful when adjusting gain particularly in a live context as adjustments can lead to feedback.

EQ Basics

EQ, or Equalisation, is a form of audio processing which allows you to adjust the volume level of a frequency or frequency range within a sound signal.

When making EQ adjustments there are three weapons in your arsenal

1. Frequency – which part of the spectrum do you want to amplify (boost) or attenuate (cut)? Measured in Hertz (Hz)
2. Gain – the amount you are boosting or cutting measured in decibels (dB)
3. 'Q' – The width of the curve which determines the range of frequencies being boosted/cut.

    Low Q = wide range of frequencies

    High Q = narrow range of frequencies

Why we use EQ:

EQ allows us to correct or change a sound to make it clearer and create the clearest possible sound. This is true both for voice and instruments.

1. Removing unneeded sound

    EQ allows us to remove the frequencies of sound that we don't need. For example, the lowest bass on a vocal signal. Leaving these low frequencies in will make the sound "muddy" and boomy. By cutting the lowest frequencies we reduce the "boominess" of the sound.
2. Selectively boosting frequencies.

    Alongside cutting unwanted frequencies we can boost the frequencies we do want to hear. For example, if we have a soloist singing with the band, we may wish to boost the frequencies in the 3-4kHz range to bring out their voice.

3. Feedback!

    Feedback is any sound engineer's nightmare and the thing which is sure to draw attention to you. Feedback occurs when the sound amplified through the speakers re-enters the microphone creating a feedback loop. This tends to start and build at one particular frequency. To eliminate feedback, we can use EQ to cut the offending frequency and thus stop it feeding back into the microphone.
    Mic placement as discussed above is also key to reducing feedback, we don't want microphones pointed at speakers which are amplifying their signal!

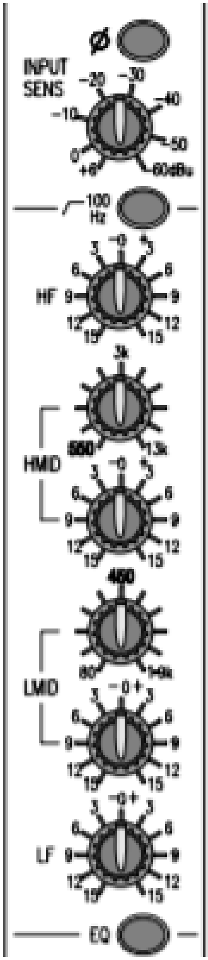
How we use EQ

Once again, listening is the most important thing we can do. When listening to the speaker's voice or to the mix of the band ask yourself if it sounds clear. *If not*, you may need to use EQ.

Tuning your ears to identify the offending frequencies is difficult and won't come naturally to everyone but with time and experience it does become easier.

Ideally, we can use the time before the service to soundcheck and make EQ adjustments but don't be afraid of making adjustments during the service. Large and fast changes will be distracting but we can make subtle changes during the service.

If you aren't sure what you're doing, join the club! Trial and error is your friend here. Don't be afraid to give things a go, over time you will learn what works and what doesn't work and you will grow in your ability to identify and solve problems more quickly.



Gain – used to adjust the input sensitivity

High frequencies (5kHz and above)

High-Mid Frequencies (550Hz-5kHz)

Low-Mid Frequencies (80-550Hz

Low Frequencies (up to 80Hz)

Note: EQ controls will look different on every sound desk but the principles are the same.

Principles of EQ

Some key principles to bear in mind when EQ-ing:

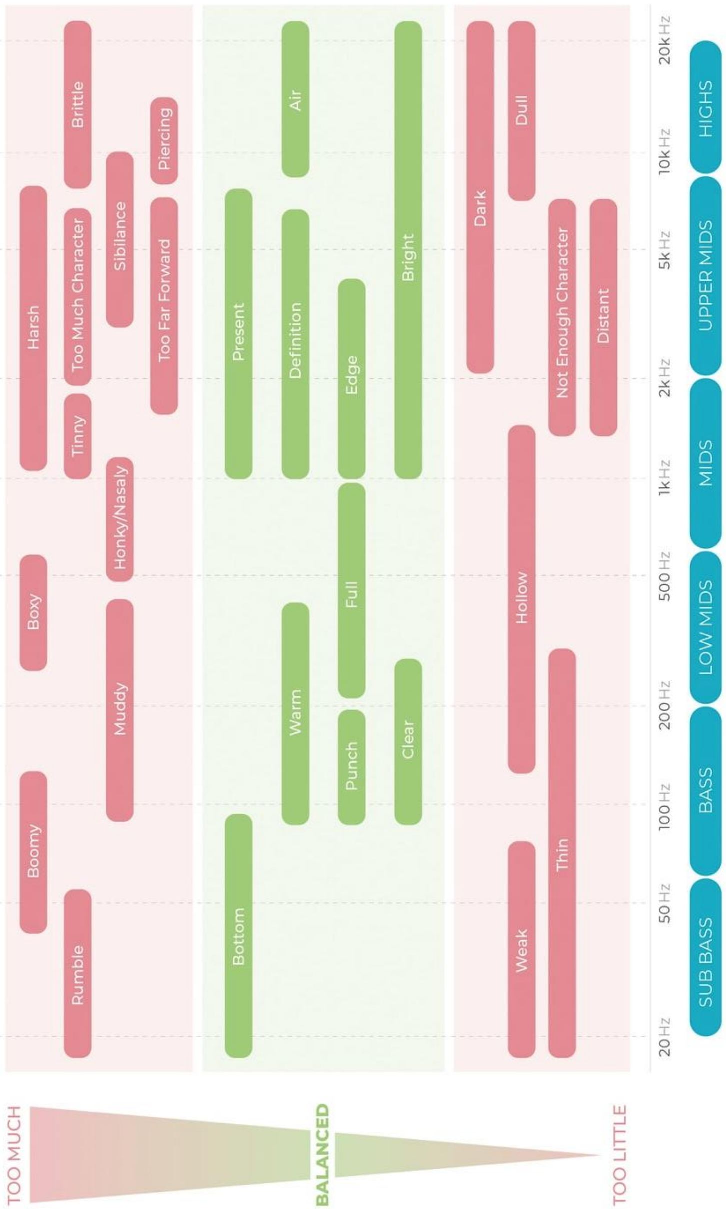
- **Use your ears, not your eyes!**
  - Most of our desks show the changes you make on the graph but the basis of any changes you make should be what you are hearing.
- Use the minimum possible amount of EQ
  - Don't make changes for the sake of changes. Changes should be motivated by what we're hearing.
- Always **cut before you boost**
  - The primary benefit of utilising EQ is to cut problematic frequencies to remove muddiness, feedback, etc. Boosting frequencies can cause us more problems rather than solving them.
- Focus on the mix
  - The end goal is always to create a **mix** which sounds good, not primarily to get every channel sounding perfect on its own. Use PAFL/CLIP to work on individual channels but always come back to listen to the mix as a whole.

Strategies for EQ

- Search and Destroy
  - Search: Narrow your channel and boost a selection of frequencies until you can hear the problem
  - Destory: Once you've found the problem frequency, 'destroy' it by cutting it.
- Cut narrow, boost wide.
  - When making cuts, cut narrow, problematic frequency ranges but when boosting, use a low Q to boost a wider frequency range.

- Make a change until you’re really sure you can hear it making a difference then dial it back slightly to ensure your adjustment is not too obtrusive.

Terminology:



EQ for Vocals and Speech

This is a rough guide for how to set up EQ for voice. This isn't a fixed formula every voice and every room is different but the basic idea remains the same

1. **High pass filter**

The first step is to cut the lowest end bass frequencies. Starting at around 80Hs we can "roll off" the bass

2. **Reduce Muddiness and Boominess**

The next step is to reduce muddiness by cutting a small notch in the 200Hs-500Hs frequency range. We don't want a massive cut but we'll use a wide Q so it's a smooth transition

3. “Brighten” vocals with a High shelf boost

In this step we want to cut the very highest frequencies over 18kHz but slightly boost high frequencies in the 5kHz-18kHz range

4. Add Prescence with a Notch Boost

Boosting the frequencies around 4kHz-5kHz range will help to bring out clarity in vocals.

5. Boost the Core Vocal Frequencies

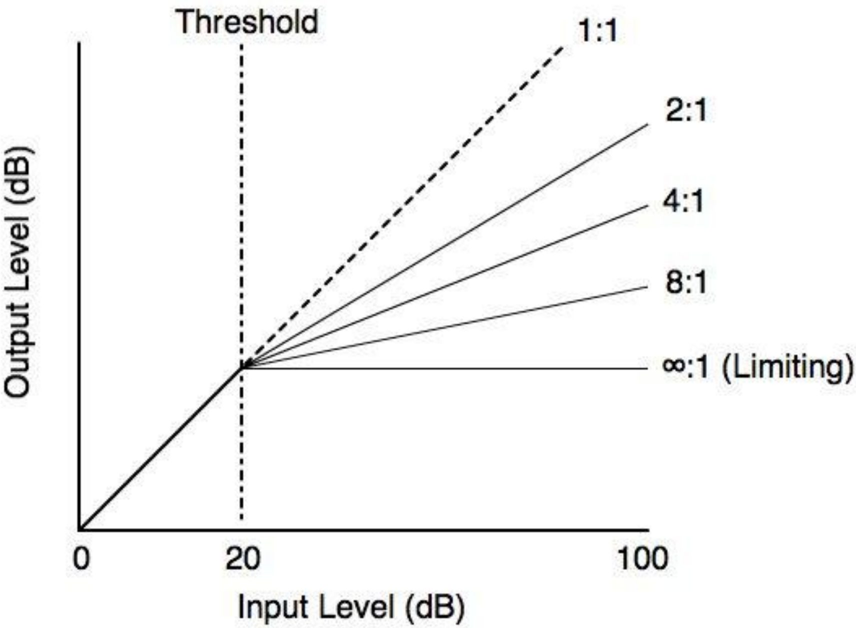
The Core vocal range is normally between 1kHz and 2kHz. Boosting this range should make the voice more clear.

6. **Putting it All Together**

   This diagram shows all of these steps put together.

Compression:

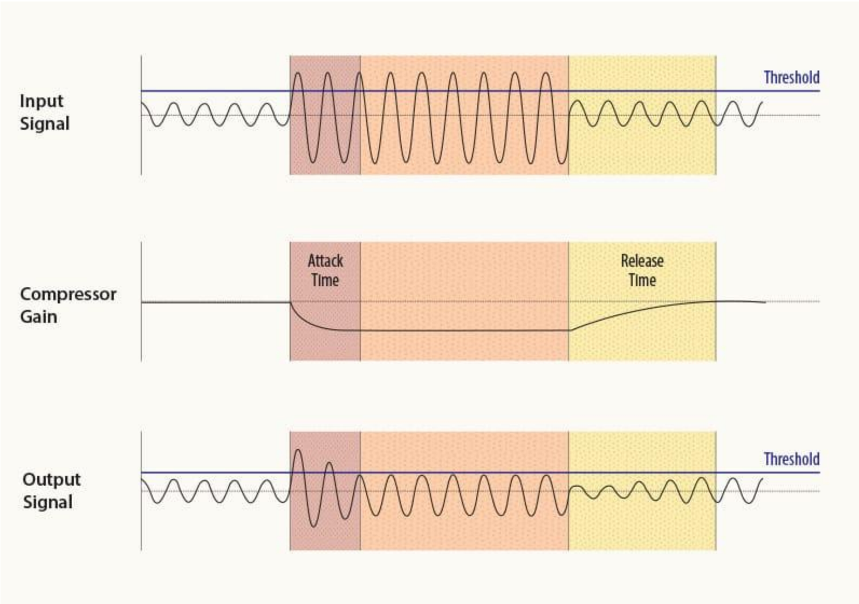
Compression is the process of reducing the dynamic range of a sound. Compression occurs when the volume level of a signal reaches a specified threshold. When this threshold is met, the sound is ‘compressed’ according to the specified ratio.



The ratio shows how loud the signal needs to be in order to allow 1 decibel to pass through the compressor.

1:1 means that for every 1dB that goes in, 1dB comes out; 2:1 means for every 2dB that goes in, 1dB comes out, and so on.

Attack and release controls affect how quickly the compressor kicks in and how quickly it is released and the sound is ‘decompressed’.



Compressors should take effect without the listeners really noticing it’s happening. Therefore, we want the attack and release to be as gradual as possible whilst also ensuring they kick in fast enough to avoid spiking in levels.

How?: Listen Well

Use Your Ears Not Just Your Eyes

If we want to create a good mix then the most important skill to develop is our ability to **listen well**.

Most of our desks show the changes you make on the graph but the basis of any changes you make should be what you are hearing.

Use the minimum possible amount of EQ

Don't make changes for the sake of changes. Changes should be motivated by what we're hearing.

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Focus on the mix

Think about the music you listen to. What sticks out most? (To test this try turning down your music until you can only just hear it, what is the last thing to go?) In almost all modern music the last things you will hear are drums and vocals which most often provide the rhythm and the melody.

These are the most important things which we want to bring out of a mix to help people to sing. They need to know the tune, and they need to keep in time.

We don't often have drums or people leading the singing in church but we do still have rhythm and melody. The rhythm will normally come from the piano and the melody from one of the other instruments.

The instrument playing the melody will change from week to week, hymn to hymn, and sometimes even from verse to verse. This means we can't just rely on setting all the instruments to the same level and leaving it there for the whole service.

Let different instruments occupy different frequency ranges.

Be Attentive to What's Going On

When you're serving, you're not eating.

It might be counter-intuitive to not be taking notes while the preacher is preaching, or not shutting your eyes when the service leader is praying, or not singing when everyone else is singing, but when you're on duty this is ok.

When preaching is going on you're not primarily listening to the message, but rather how the speech is sounding. It would be as bizarre as your waiter at a restaurant sitting down to eat when they should be bringing you a meal! This is how we love our neighbour.